Investigating Magnitude and Epoch-Dependent Sampling in TrivialAugment

Submitted by

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Project Completion Certificate

Yet to be filled.

Acknowledgement

I would like to express my sincere gratitude to my supervisor for giving me the opportunity to undertake this research project and for introducing me to data augmentation.

I am particularly grateful for the guidance provided throughout the project. I also appreciate the computational resources and support that made the repeated training experiments possible. The project gave me the opportunity to move beyond simply implementing a published method and to investigate, through experimentation, whether some modifications to its sampling procedure could actually improve performance.

Finally, I would like to thank everyone who contributed directly or indirectly to the completion of this project.

Aim/Objectives

The primary aim of this project was to investigate whether the performance of TrivialAugment could be improved by changing how augmentation operations and their magnitudes are sampled.

The specific objectives were:

1. To study and implement TrivialAugment.
2. To establish a TrivialAugment baseline on CIFAR-10.
3. To investigate whether applying more than one augmentation operation can improve generalization.
4. To investigate whether the magnitude range should depend on the number of operations applied.
5. To investigate whether magnitude should change as training progresses.
6. To investigate whether the number of operations should itself change with training epoch.
7. To compare these sampling strategies against the original TrivialAugment baseline.
8. To examine whether the most developed strategy transfers to a second dataset, STL-10.

The central research question was:

> *Can TrivialAugment be improved by changing how augmentation magnitudes and augmentation operations are sampled?*

The initial hypothesis was that progressively increasing augmentation strength could improve generalization. The intuition was that the model could first learn from relatively subtle image variations and later be exposed to increasingly stronger transformations.

Introduction

Data augmentation is a standard technique in image classification in which transformations are applied to training images in order to increase the diversity of the examples seen by a model. When the transformations preserve the semantic class of an image, augmentation can encourage a model to learn features that are less dependent on incidental characteristics of the training data.

A central difficulty in data augmentation is deciding how transformations should be selected and how strongly they should be applied. A transformation that is too weak may introduce little useful diversity, while a transformation that is too strong may distort information that is important. The number of transformations applied to a single image introduces another dimension of this trade-off.

TrivialAugment was selected as the starting point of this project because of its simplicity. Its sampling procedure made it particularly suitable for controlled modifications. Rather than designing new augmentation transformations, the project could investigate whether the *sampling policy* itself could be improved.

The project therefore kept the augmentation functions fixed and changed only the way in which the number of operations and their magnitudes were sampled. The investigation developed through three major experimental families.

The first family varied the number of augmentation operations and magnitude ranges. The second family made the maximum magnitude dependent on the training epoch. The third family extended the second by making the probability of selecting one or two operations dependent on the epoch through a sigmoid.

A key idea behind the first family was that applying more operations does not necessarily require keeping the same maximum magnitude for every operation. For example, the choices of maximum magnitude 30, 15, and 10 for one, two, and three operations respectively were motivated by the qualitative heuristic

$$1(30) \approx 2(15) \approx 3(10),$$

which was treated only as a rough proxy for total augmentation strength.

The later experiments investigated a different qualitative idea: instead of using a stationary augmentation distribution throughout training, the distribution could gradually change. Initially, relatively mild augmentation could be emphasized, while stronger augmentation could become more common later.

The sigmoid schedule was chosen because it provides a simple way of making this transition gradual. Different scales of the sigmoid were then tested to determine whether the rate at which the probability changed affected performance.

Theoretical Framework of Topic

Data Augmentation and Generalization

Let x denote an input image with class label y. An augmentation transformation with operation T and magnitude m produces

$$x' = T(x; m).$$

For a useful augmentation, x' should remain semantically consistent with x while introducing variation.

This gives rise to a basic trade-off. If the transformation is too weak, the additional image diversity may be limited. If it is too strong, the transformed image may no longer preserve the information needed for classification. Therefore, greater augmentation strength cannot automatically be assumed to produce better generalization.

TrivialAugment

The augmentation pool used in this project consisted of fourteen operations:

$$\begin{aligned}\mathcal{A} = \{&\text{ShearX, ShearY, TranslateX, TranslateY, Rotate,}\\ &\text{AutoContrast, Equalize, Invert, Solarize, Posterize,}\\ &\text{Contrast, Colour, Brightness, Sharpness}\}.\end{aligned}$$

The augmentation functions themselves were not modified during the project. Only the sampling procedure was changed.

In the baseline implementation, one operation was selected randomly and its magnitude was sampled uniformly from the available range. For the CIFAR-10 baseline this was

$$m \sim U\{0, 1, \ldots, 30\}.$$

When multiple operations were used, the transformed image could be written as

$$x_0 = x, \qquad x_i = T_i(x_{i-1}; m_i), \quad i = 1, \ldots, N.$$

Operations were sampled independently and with replacement, so the same operation could be selected more than once. Magnitudes were sampled independently for the selected operations.

Varying the Number of Operations

The first experimental family investigated whether increasing the number of operations could improve augmentation.

For two possible operation counts,

$$N \sim U\{1, 2\}.$$

For three possible operation counts,

$$N \sim U\{1, 2, 3\}.$$

The primary experiments used maximum magnitudes of 15 for two-operation sampling and 10 for three-operation sampling. The qualitative reasoning was that if multiple operations are applied, their individual magnitudes might need to be reduced to avoid producing an excessively strong combined transformation.

Epoch-Dependent Magnitude

The second experimental family made the maximum magnitude a function of training epoch. Let E denote the current epoch and $E_{\max}$ the total number of epochs. The main linear schedule was

$$M_{\max}(E) = \text{round}\left(1 + \frac{29E}{E_{\max}}\right).$$

Thus, over a 200-epoch run, the maximum magnitude increases from 1 at the start of training to 30 at the final epoch.

With a fixed minimum magnitude of zero,

$$m \sim U\{0, \ldots, M_{\max}(E)\}.$$

A second class of schedules also moved the lower endpoint:

$$M_{\min}(E) = \max\{0, M_{\max}(E) - c\},$$

followed by

$$m \sim U\{M_{\min}(E), \ldots, M_{\max}(E)\}.$$

The constants tested included $c = 15$ for the 30-scale maximum, and $c = 7$ for the 15-scale maximum.

Sigmoid Scheduling of Operation Count

The third and final experimental family extended the epoch-dependent magnitude experiment. The magnitude was still sampled uniformly using the linear epoch-dependent schedule, while the probability of selecting one or two operations was made epoch-dependent.

For the sigmoid experiments,

$$N \in \{1, 2\}.$$

The code defined

$$x(E) = 6\frac{E}{E_{\max}} - 3$$

for the Large schedule. The probability assigned to one operation was

$$p_1(E) = \frac{1}{1 + e^{x(E)}},$$

while the probability of two operations was

$$p_2(E) = 1 - p_1(E).$$

Thus, approximately, the beginning of training strongly favours one operation, the middle of training approaches a 50–50 split, and the end of training strongly favours two operations.

Three schedules were studied:

$$x_{\text{Small}}(E) = 2\frac{E}{E_{\max}} - 1,$$

$$x_{\text{Medium}}(E) = 4\frac{E}{E_{\max}} - 2,$$

and

$$x_{\text{Large}}(E) = 6\frac{E}{E_{\max}} - 3.$$

The larger scale produces a more strongly skewed probability distribution away from the midpoint, while the smaller scale produces a more gradual transition.

For these sigmoid experiments, magnitude was sampled uniformly from

$$m \sim U\{0, \ldots, M_{\max}(E)\},$$

using the same 1–30 linear maximum-magnitude schedule as the preceding epoch-dependent experiment.

Methodology Followed

Overall Experimental Design

The investigation followed a progressive structure:

1. Implement the original TrivialAugment method and establish a CIFAR-10 baseline.
2. Vary the number of augmentation operations and investigate corresponding magnitude ranges.
3. Introduce a linear epoch-dependent maximum magnitude.
4. Investigate schedules in which the minimum magnitude also changes with epoch.
5. Extend the epoch-dependent magnitude approach by making operation count epoch-dependent through a sigmoid probability.
6. Evaluate the sigmoid strategy on STL-10 in addition to CIFAR-10.

Dataset and Preprocessing

CIFAR-10 was selected as the primary dataset because it is a standard image classification benchmark and is computationally manageable for repeated experimentation. The standard train/test split was used.

The CIFAR-10 training pipeline applied random cropping with padding, the investigated augmentation method, tensor conversion, and dataset-specific normalization. The CIFAR-10 normalization used in the implementation was

$$\mu = (0.4914, 0.4822, 0.4465),$$

$$\sigma = (0.2470, 0.2435, 0.2616).$$

STL-10 was used as a second dataset to test whether the sigmoid strategy showed similar behaviour outside CIFAR-10. The standard train/test split and dataset-specific preprocessing were used.

Oxford Flowers 102 was also explored during the project. Images were resized to 224×224 and the necessary changes were made to the classification setup. However, these results are excluded from the principal analysis because the resulting accuracies were low and the experiments were not sufficiently repeated to support a meaningful conclusion.

Model and Training Configuration

The principal experiments used WideResNet-28-10 and were trained from scratch. The main CIFAR-10 and sigmoid experimental configuration was:

Table 1: Training configuration used in the principal experiments.

Architecture	WideResNet-28-10
Epochs	200
Batch size	128
Optimizer	Nesterov SGD
Learning rate	0.1
Momentum	0.9
Weight decay	5×10^{-4}
Learning-rate scheduler	CosineAnnealingLR
Loss function	Cross-entropy
Dropout rate	0.3
Initialization	From scratch

For the Oxford Flowers experiments, the batch size was changed to 32 and the input resolution was adjusted to 224×224.

The same principal training configuration was retained across the CIFAR-10 experiments so that the augmentation sampling strategy remained the main experimental variable.

Baseline

The baseline consisted of the original TrivialAugment sampling procedure: one randomly selected operation and a uniformly sampled magnitude. The baseline was evaluated using four CIFAR-10 runs.

First Experimental Family: Number of Operations

The primary first-family experiments were:

Table 2: First experimental family on CIFAR-10.

Configuration	Runs	Average best test accuracy
$N = \{1, 2\}$, $M = [0, 15]$ for all N	3	96.59%
$N = \{1, 2\}$, $M = [0, 30]$ if $N = 1$, $[0, 15]$ if $N = 2$	3	96.79%
$N = \{1, 2, 3\}$, $M = [0, 10]$ for all N	1	96.21%
$N = \{1, 2, 3\}$, $M = [0, 30]$ if $N = 1$, $[0, 15]$ if $N = 2$, $[0, 10]$ if $N = 3$	1	96.36%
$N = \{1, 2, 3, 4\}$, $M = [0, 7]$ for all N	1	94.69%

The four-operation experiment was not pursued further because it showed poor training behaviour and unstable convergence.

Second Experimental Family: Epoch-Dependent Magnitude

The second family investigated a linear change in magnitude over training.

Table 3: Second experimental family on CIFAR-10.

Configuration	Runs	Average best test accuracy
$N = \{1, 2\}$, $M_{\max} = 1 + \frac{29E}{E_{\max}}$, $M_{\min} = 0$	1	96.65%
Same $M_{\max}$, $M_{\min} = \max(0, M_{\max} - 15)$	1	96.75%
$N = \{1, 2\}$, $M_{\max} = 1 + \frac{14E}{E_{\max}}$, $M_{\min} = 0$	1	96.45%
Same 15-scale maximum, $M_{\min} = \max(0, M_{\max} - 7)$	1	96.60%

Third Experimental Family: Sigmoid Operation Count

The third family kept the linear epoch-dependent magnitude schedule and changed the operation-count distribution.

Table 4: Sigmoid experiments on CIFAR-10.

Configuration	Runs	Average best test accuracy
Sigmoid Large	5	96.83%
Sigmoid Medium	5	96.74%
Sigmoid Small	5	96.70%

Evaluation and Repetition

For each run, the highest test accuracy reached during training was recorded. When multiple runs were available, the reported value was the mean of the per-run best test accuracies.

The CIFAR-10 TrivialAugment baseline was run four times. The sigmoid variants were each run five times. Several exploratory configurations were run once, while more promising configurations received repeated runs. Approximately 40 training runs were conducted in total.

A random seed was used during the experiments. The seed was changed between repeated runs.

Analysis and Interpretation of Research Project

Baseline Performance

The TrivialAugment baseline achieved

$$\boxed{96.99\%}$$

average best test accuracy on CIFAR-10 across four runs.

This is an important result because it establishes that the starting method was already highly effective. Consequently, the proposed modifications were being evaluated against a strong baseline rather than against an underperforming implementation.

Effect of Increasing the Number of Operations

The first experimental family did not support the hypothesis that simply using more augmentation operations would improve generalization.

For $N = \{1, 2\}$ with a common range of $[0, 15]$, the average best test accuracy was 96.59%. Allowing one operation to use $[0, 30]$ and two operations to use $[0, 15]$ improved this to 96.79%, but this was still 0.20 percentage points below the TrivialAugment baseline.

The three-operation configurations were weaker. The uniform $[0, 10]$ configuration achieved 96.21%, while the operation-dependent magnitude configuration achieved 96.36%. The four-operation experiment performed substantially worse at 94.69% and was accompanied by poor training accuracy and unstable convergence.

These observations indicate that increasing the number of operations is not sufficient to improve performance. In fact, applying too many operations can make the augmentation excessively disruptive.

The results also suggest that the distribution of magnitude matters. The two-operation configuration in which the magnitude range depended on the number of sampled operations performed better than the version using the same $[0, 15]$ range for both operation counts. However, neither exceeded the original baseline.

Effect of Epoch-Dependent Magnitude

The second experimental family tested the central hypothesis that augmentation should become progressively stronger during training.

The 1–30 maximum schedule with minimum magnitude zero achieved 96.65%. Changing the minimum magnitude produced slightly higher results, with the best configuration reaching 96.79%. The lower maximum-magnitude schedule, which increased only from 1 to 15, achieved 96.45% with minimum zero and 96.60% when the minimum was also increased.

None of these configurations exceeded the 96.99% baseline.

The results therefore do not support the claim that a simple linear increase in augmentation magnitude necessarily improves generalization. They also suggest that the lower end of the magnitude distribution can matter: two schedules with the same maximum magnitude produced different accuracies when their minimum magnitudes were changed.

Because the configurations in this family were mostly evaluated in single runs, small differences such as 96.75% versus 96.79% should be interpreted as exploratory rather than as definitive evidence that one schedule is better.

Sigmoid Scheduling of Operation Count

The third experimental family extended the epoch-dependent magnitude experiment by changing the probability of selecting one or two operations over training.

The Large sigmoid configuration achieved 96.83%, the Medium configuration 96.74%, and the Small configuration 96.70%. The Large configuration was the strongest of the three, but it remained 0.16 percentage points below the 96.99% baseline.

This result is nevertheless useful because each sigmoid variant was repeated five times, providing more repeated evidence than most of the exploratory configurations in the earlier families.

The experiments therefore demonstrate that sigmoid scheduling provides a simple mech-

anism for controlling augmentation complexity over training, but they do not establish that this mechanism improves upon TrivialAugment.

STL-10 Evaluation

The sigmoid variants were additionally evaluated on STL-10.

Table 5: STL-10 results.

Configuration	Runs	Average best test accuracy
TrivialAugment	2	84.93%
Sigmoid Large	2	83.92%
Sigmoid Medium	2	83.67%
Sigmoid Small	2	83.38%

The TrivialAugment baseline outperformed all three sigmoid variants on STL-10. The best sigmoid result was 83.92%, which was 1.01 percentage points below the baseline.

Thus, the epoch-dependent sigmoid operation-count schedule did not show a consistent advantage across the two principal datasets.

Interpretation of the Hypothesis

The initial hypothesis was that progressively increasing augmentation strength would improve generalization. The experimental evidence obtained in this project does not support that hypothesis as a general rule.

One possible interpretation is that the stationary randomness of TrivialAugment is already beneficial. Because the original method samples transformations and magnitudes randomly throughout training, the model is already exposed to a broad range of augmentation strengths. Imposing a specific temporal schedule may therefore fail to add useful information.

A second interpretation is that augmentation has a trade-off. Moderate transformations can increase data diversity, while overly strong or overly complex transformations can make examples less useful for learning. The poor behaviour observed for four operations is consistent with this possibility.

A third possibility is that the usefulness of an augmentation schedule depends on the

interaction between dataset, architecture, optimization, and augmentation operations. The present experiments are not sufficient to identify the precise mechanism.

Most importantly, the results demonstrate that an intuitive research hypothesis should not be assumed to hold merely because it is qualitatively reasonable. In this case, the experiments provided evidence against the simple idea that progressively increasing augmentation strength is automatically beneficial.

Conclusion & Suggestion of Research Project

This project investigated whether TrivialAugment could be improved by changing how augmentation operations and magnitudes are sampled.

The principal quantitative conclusion is that none of the investigated modifications consistently improved upon the original TrivialAugment baseline on the two principal datasets.

On CIFAR-10, the baseline achieved 96.99%, while the strongest investigated configuration achieved 96.83%. On STL-10, the baseline achieved 84.93%, while the strongest sigmoid configuration achieved 83.92%.

The original hypothesis was therefore not confirmed. In particular, the experiments do not support the claim that progressively increasing augmentation strength is inherently beneficial.

Nevertheless, the project produced several useful observations. First, increasing the number of operations too aggressively can be harmful. Second, the magnitude distribution matters, not merely the number of operations. Third, epoch-dependent schedules can be implemented in a simple and controlled manner, but their intuitive appeal does not guarantee improved generalization. Finally, the experiments highlight the difficulty of improving a strong and simple baseline such as TrivialAugment through straightforward modifications to its sampling mechanism.

Future work could investigate more datasets, more independent random seeds, additional architectures, a wider range of augmentation policies, and more systematic statistical analysis. A stronger theoretical treatment of what constitutes total augmentation strength would also be valuable. Another possible direction would be to make the augmentation schedule adaptive to model behaviour or training dynamics rather than dependent only on epoch.

Results Achieved

Table 6: Summary of principal results.

Dataset	Configuration	Average best test accuracy
CIFAR-10	TrivialAugment baseline	96.99%
CIFAR-10	Best first-family configuration	96.79%
CIFAR-10	Best second-family configuration	96.75%
CIFAR-10	Sigmoid Large	96.83%
STL-10	TrivialAugment baseline	84.93%
STL-10	Sigmoid Large	83.92%

The best result in the first family was 96.79%, obtained using $N = \{1, 2\}$ with magnitude sampled from $[0, 30]$ when one operation was selected and $[0, 15]$ when two operations were selected.

The best result in the second family was 96.75%, obtained using the 1–30 linear maximum-magnitude schedule with a moving minimum magnitude

$$M_{\min} = \max(0, M_{\max} - 15).$$

The best sigmoid result on CIFAR-10 was the Large configuration at 96.83%. On STL-10, the Large configuration was also the strongest sigmoid variant, at 83.92%, but remained below the TrivialAugment baseline.

Approximately 40 training runs were conducted over the course of the investigation. The overall result was that the original TrivialAugment baseline remained the strongest method among the configurations tested on the principal datasets.

References

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Declaration on the Use of Artificial Intelligence

Artificial intelligence tools were used as writing assistance in the preparation of this report. The tools were used primarily for organizing, drafting, and refining the presentation of material based on the author's research, experiments, and analysis. The experimental design, implementation, results, interpretation, and conclusions are the author's own work. The author takes full responsibility for the accuracy and content of this report.